

Single-electron transistors in heterostructure nanowires

Key achievement:

- A controlled, reproducible SET is created using a **bottom-up nanowire growth method**, rather than traditional lithography.

What is traditional lithography ?

Traditional lithography, primarily optical photolithography, is generally considered unsuitable for fabricating Single Electron Transistors (SETs) because it cannot achieve the ultra-small features—typically under 10 nm—required for the device to operate at room temperature. While standard industrial optical lithography is limited to feature sizes above 100 nm.

10 nm threshold refers specifically to the size of the "island"—the small conductive region at the heart of the transistor.

The Island: A tiny conductive dot (usually <10nm) separated from the rest of the circuit by ultra-thin insulators.

Why 10 nm?

The transistor relies on a phenomenon called the **Coulomb Blockade**. For this to work at room temperature, the energy required to add a single electron to the island (charging energy) must be much higher than the surrounding thermal energy.

The charging energy ($E_c = e^2 / C$) must be greater than the surrounding thermal energy ($k_B T$) in single-electron transistors (SETs) to prevent random thermal fluctuations from causing electrons to tunnel onto or off the island, ensuring that electron transport is controlled solely by the gate voltage.

- Smaller is better: As the island gets smaller, its capacitance decreases, which increases the energy needed to push an electron onto it.
- The limit: If the island is larger than 10 nm, the charging energy is too low, and random thermal noise will "knock" electrons onto the island, breaking the transistor's control.

Coulomb Blockade is a physical phenomenon where the addition of a single electron to a tiny "island" is blocked by the electrical repulsion of the electrons already there.

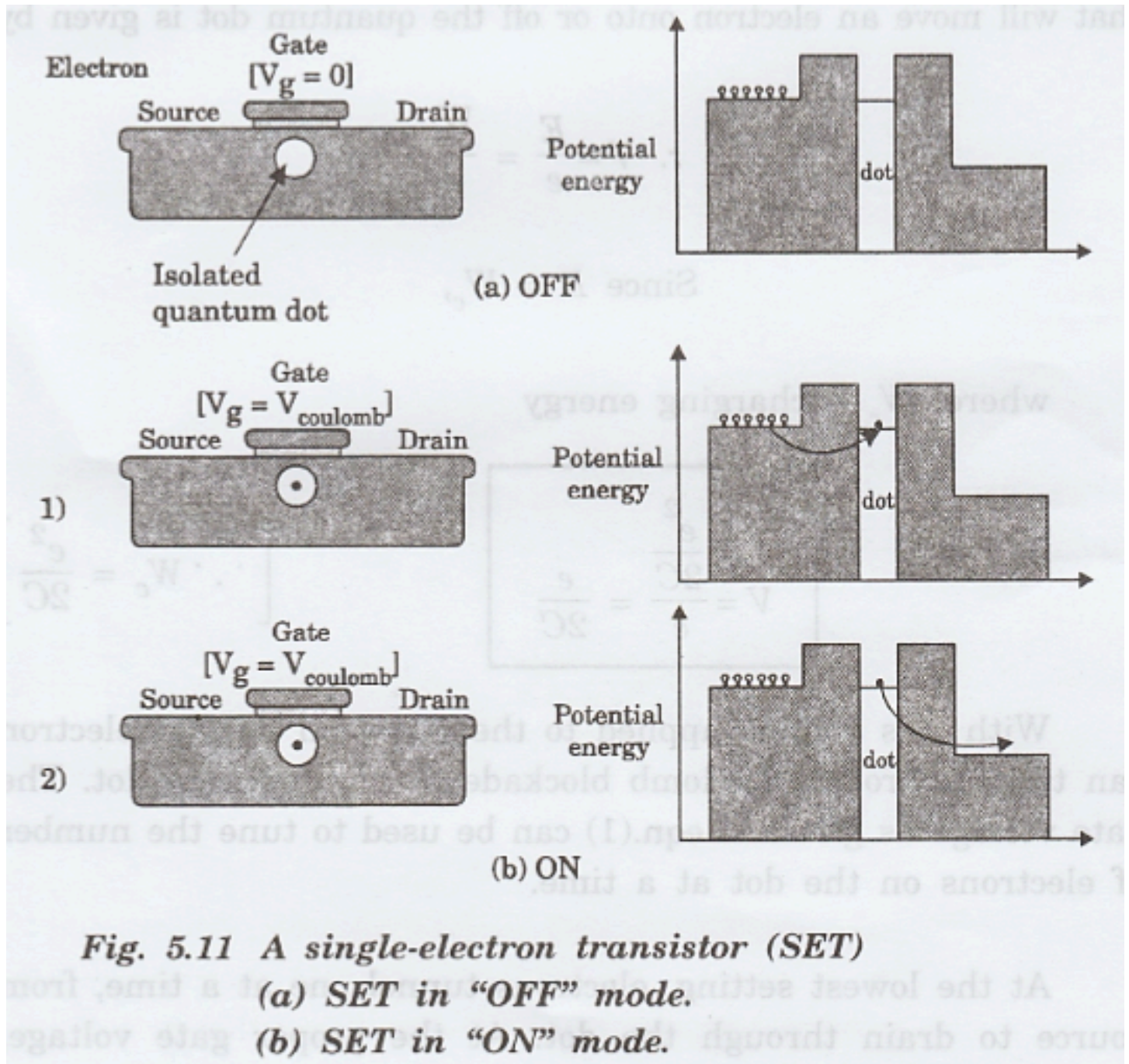
Think of it like a **crowded elevator**:

1. The elevator (the "island") is so small that if one person (an electron) is inside, there is no physical room for a second person.
2. To force a second person in, you would have to push incredibly hard to overcome the resistance.

3. In a transistor, this "push" is the **voltage**. Unless you provide enough voltage to overcome this repulsion, no current can flow.

The **bottom-up nanowire growth method** for a **Single Electron Transistor (SET)** involves building the transistor atom-by-atom or layer-by-layer, rather than carving it out of a larger block.

This method is highly effective because it allows for the precise creation of a "**quantum dot**"—a tiny conducting island where only one electron can reside at a time



The paper demonstrates how to **fabricate and study single-electron transistors (SETs)** using **semiconductor nanowires** with **engineered heterostructures (InAs/InP)**.

Definition:

A SET is a device where **electrons pass one at a time** through a small conducting region called an **island**.

Structure:

- Two leads (source and drain)
- A small island between them
- Tunnel barriers separating the island from leads
- A gate electrode controlling charge

Working Principle:

- Electron transport is governed by **Coulomb blockade**
- Current flows only when energy conditions allow adding one electron

Important condition:

$$k_B T \ll \frac{e^2}{C}$$

- $k_B T$: thermal energy
- e^2/C : charging energy

The energy diagram of a Single Electron Transistor (SET) illustrates how the **Coulomb blockade** effect controls the movement of individual electrons between a source and drain through a central **quantum dot** (island)

Energy Band Diagram will be like :

Source and Drain: These are electrodes with a continuous range of filled energy levels up to their respective Fermi levels

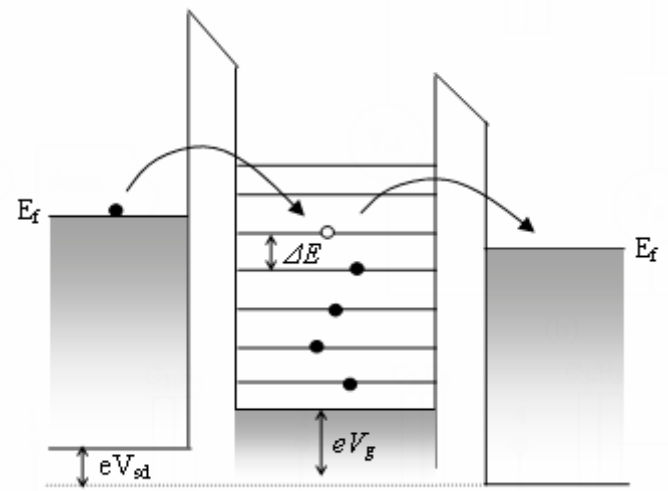
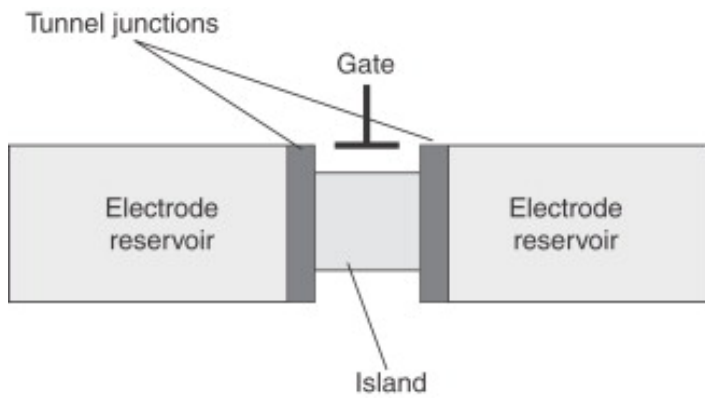
Quantum Dot (Island): A small conducting region between two tunnel barriers. Unlike the bulk electrodes, the island has **discrete energy levels** due to quantum confinement.

Charging Energy (E_c): This is the electrostatic energy required to add a single electron to the island, defined as $E_c = e^2 / 2C$, where C is the total capacitance of the island.

The Role of Gate Voltage (V_g)

The gate electrode is capacitively coupled to the island. By adjusting V_g you can "push" or "pull" the discrete energy levels of the island up or down.

In a **Single Electron Transistor (SET)**, the energy levels of the central "island" (quantum dot) change dynamically as electrons enter and leave. This behavior is governed by the **Charging Energy** $E_c = e^2 / 2C$, which is the energy cost to add one additional electron to the island.



1. No Electrons (Neutral State)

- **Band Diagram:** The discrete energy levels of the island are positioned such that no level falls between the Fermi levels of the source and drain.
- **Electron Motion:** Current is **blocked**. Although an electron in the source "wants" to move to the drain, there is no available state at that energy level on the island. The energy required to add an electron E_c is higher than the potential energy available from the source.

2. One Electron Passes (Tunneling On)

- **Band Diagram:** By applying a **Gate Voltage** V_g the island's energy levels are shifted downward until a specific level E_n aligns with the source Fermi level
- **Electron Motion:** A single electron from the source can now "tunnel" through the barrier onto the island. Once it is there, the island's total energy immediately **increases by E_c**
- **Exit:** This new energy state allows the electron to tunnel again, this time off the island and into the lower-energy drain.

3. Second Electron (Coulomb Blockade)

- **Band Diagram:** As soon as the first electron occupies the island, the energy levels for a *second* electron E_{n+1} shift upward by E_c .
- **Electron Motion:** Even if V_g is constant, the second electron is **blocked** from entering. It would need an additional E_c worth of energy to overcome the electrostatic repulsion from the electron already sitting on the island.
- **Sequential Flow:** The second electron can only enter *after* the first one has hopped off into the drain, resetting the island's energy. This results in the "one-by-one" electron motion characteristic of SETs.

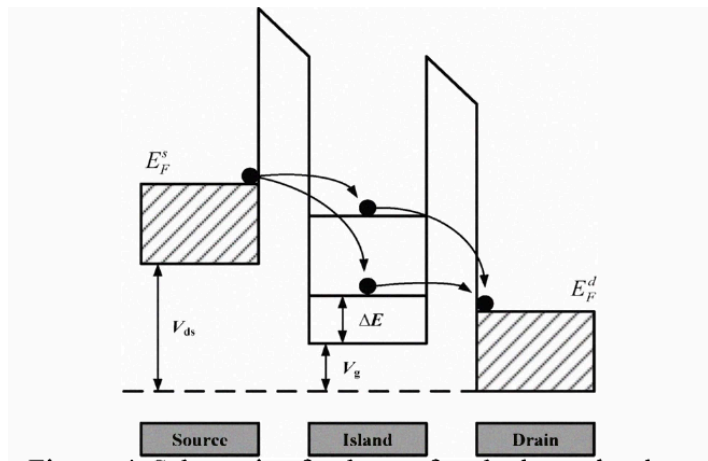
when second electron enters the energy is provided again by changing the gate voltage right

Yes, exactly. To get a **second** electron onto the island while the first one is still there, you have to "pay" the energy penalty.

Here is how it works in the diagram:

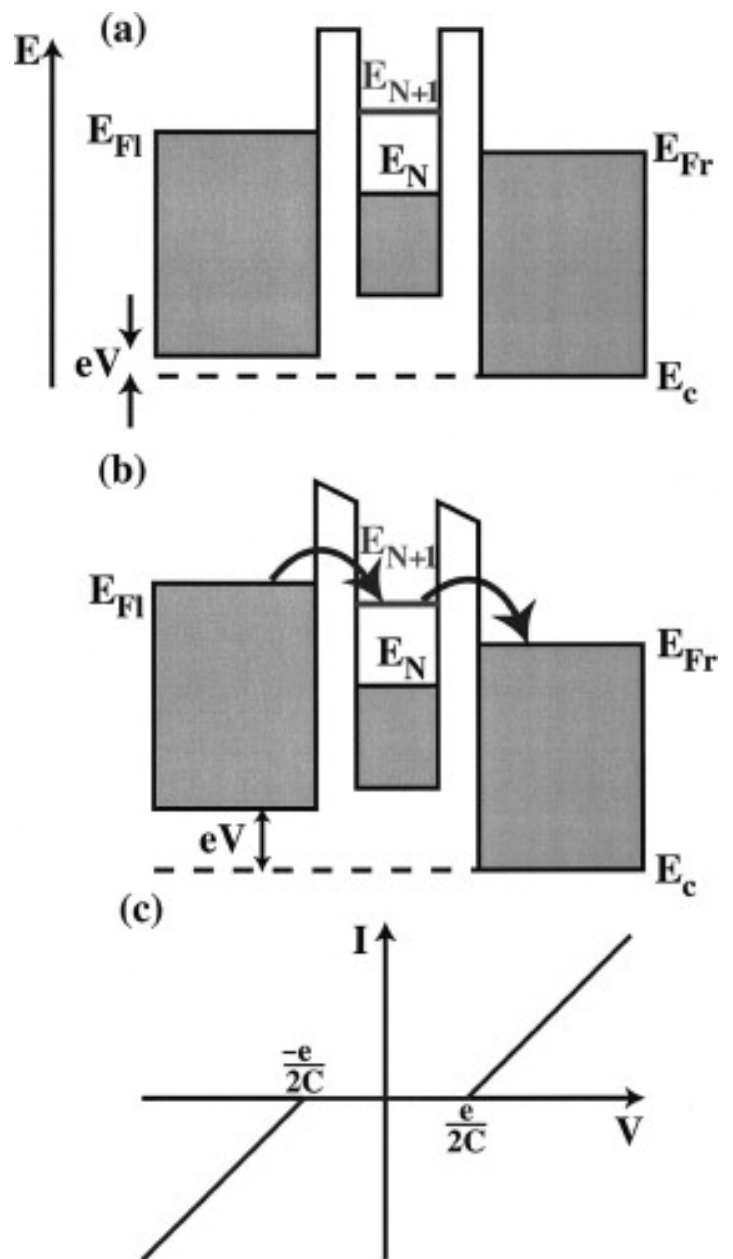
1. **The First Electron stays:** When the first electron enters, the island's electrostatic potential rises by E_c . This pushes all the energy levels **up**, moving the next empty level ($N + 1$) above the source's Fermi level. The second electron is now blocked.
2. **Changing V_g :** To overcome this, you must increase the **Gate Voltage** (V_g). This applies an external electric field that "pulls" the island's energy levels back **down**.
3. **The Second Electron enters:** Once you've increased V_g enough to counteract that E_c jump, the next discrete energy level aligns with the source again. Now, a second electron has enough energy to tunnel onto the island.

In short: the gate voltage acts like a **lever**. Every time an electron enters and "lifts" the energy levels, you have to push the lever (increase V_g) further down to let the next one in.



$$E_{n_x, n_y, n_z} = \frac{\hbar^2}{8m^*L^2} (n_x^2 + n_y^2 + n_z^2)$$

$$E_{n,l} = \frac{\hbar^2 \chi_{n,l}^2}{2m^*R^2}$$



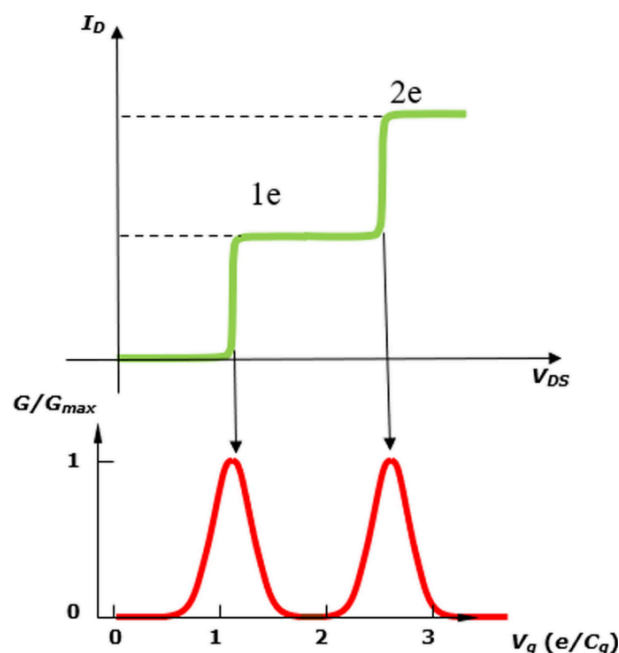
The Mechanism

- **Diagram (a) - Blockade State:** The energy level E_{N+1} of the quantum dot lies above the Fermi level (E_{FI}) of the source. Even if a small bias voltage (V) is applied, no electron can tunnel into the dot because it lacks sufficient energy to overcome the electrostatic repulsion of the N electrons already present. No current flows.
- **Diagram (b) - Tunneling State:** As the bias voltage increases, the source's Fermi level is raised relative to the dot. Once E_{FI} reaches the discrete energy level E_{N+1} , an electron can tunnel into the dot and then out to the drain (E_{FD}), enabling current flow.

The I-V Characteristic (c)

The bottom graph shows the resulting current-voltage relationship:

- **Coulomb Gap:** Near zero voltage, there is a "dead zone" where current is zero. This happens between $-\frac{e}{2C}$ and $\frac{e}{2C}$, where e is the elementary charge and C is the capacitance.
- **Charging Energy:** To add an electron to the island, the system must overcome the charging energy $E_c = \frac{e^2}{2C}$.
- **Linear Growth:** Once the voltage exceeds this threshold, the blockade is lifted, and current increases linearly with voltage.



1. The Coulomb Staircase (I_D vs. V_{DS})

The top graph displays the drain current (I_D) as a function of the drain-source voltage (V_{DS}).

- **The "Staircase" Effect:** As V_{DS} increases, the current does not rise linearly. Instead, it moves in discrete steps labeled **1e** and **2e**.
- **Physics:** Each step represents the addition of exactly one electron to the "island" (the central part of the transistor). Current is blocked until the voltage provides enough energy to overcome the electrostatic charging energy ($E_c = e^2/2C$) required to add an extra electron.

2. Coulomb Oscillations (G/G_{max} vs. V_g)

The bottom graph shows the normalized conductance (G/G_{max}) versus the gate voltage (V_g), which is scaled by e/C_g .

- **Conductance Peaks:** The red peaks indicate specific gate voltages where the energy levels of the island align with the source and drain leads, allowing electrons to tunnel through one by one.
- **The Blockade Region:** Between the peaks, the conductance drops to zero. This is the **Coulomb Blockade** regime, where the number of electrons on the island is stable (e.g., N or $N + 1$), and the energy cost to add or remove an electron is too high for current to flow.

1. Top Graph: Coulomb Staircase (I_D vs. V_{DS})

The green curve shows the drain current (I_D) as a function of the source-drain bias voltage (V_{DS}).

- **Staircase Shape:** Current increases in discrete steps rather than a smooth curve. Each step corresponds to adding exactly one more electron to the "island" or quantum dot.
- **Plateaus (1e, 2e):** The flat regions represent states where the number of electrons on the island is stable. To add another electron, the bias voltage must overcome the **charging energy** ($E_C = e^2/2C_\Sigma$), resulting in the next jump.
- **Coulomb Blockade:** Near $V_{DS} = 0$, the current is suppressed because the energy required to add a single electron is higher than the available thermal or electrical energy.

2. Bottom Graph: Coulomb Oscillations (G/G_{max} vs. V_g)

The red curve shows the normalized conductance (G) as a function of the gate voltage (V_g), which is scaled by e/C_g .

- **Conductance Peaks:** These peaks occur at specific gate voltages where the energy levels of the island align with the chemical potentials of the source and drain. At these points, electrons can tunnel on and off the island freely.
- **Relationship to Staircase:** The arrows indicate that the conductance peaks align with the **vertical transitions** in the I_D staircase. While the staircase shows how bias pulls electrons through, the gate voltage peaks show the "resonant" points where the island's charge state can change by one electron.
- **Spacing:** The peaks are periodic, with a spacing of approximately $\Delta V_g = e/C_g$.

Summary Table

Feature	Top Graph (I_D vs. V_{DS})	Bottom Graph (G vs. V_g)
Phenomenon	Coulomb Staircase	Coulomb Oscillations
Controlled By	Bias Voltage	Gate Voltage
Key Insight	Discrete electron addition	Resonant tunneling points
Unit Scale	$1e, 2e$ (Charge states)	e/C_g (Gate capacitance)

Proposed Solution: Heterostructure Nanowires

Materials used:

- **InAs (Indium Arsenide)** → conducting region (island)
- **InP (Indium Phosphide)** → tunnel barriers

Structure:

- InP–InAs–InP → forms a **double barrier**
- InAs region acts as **quantum dot (island)**

Dimensions:

- Diameter: 55–65 nm
- Island length: 80–100 nm

- Barrier thickness: ~5 nm

Because the **InAs** has a smaller bandgap than the **InP** barriers, it creates a “Quantum Dot” that traps electrons.

5. Fabrication Method

Step-by-step:

1. **Nanowire growth**
 - Method: chemical beam epitaxy
 - Seed: gold nanoparticles
2. **Heterostructure formation**
 - Insert InP barriers during growth
3. **Transfer wires**
 - Onto Si substrate with SiO₂ layer
4. **Electrode fabrication**
 - Electron-beam lithography
 - Contacts: Ni + Au
5. **Surface treatment**
 - Etching with (NH₄)₂S_x to improve contacts

6. Control Experiment (Reference Wires)

Homogeneous InAs nanowires:

- No InP barriers
- Observations:
 - Linear I–V (Ohmic behavior)
 - No Coulomb blockade
 - Resistance increases for thinner wires

7. Electrical Measurements (Main Results)

7.1 Coulomb Blockade

Observed:

- Current is **blocked at low voltage**
- Only flows when gate voltage is tuned

Charging energy:

- ~4 meV (for 55 nm wire)

7.2 Conductance Oscillations

Key observation:

- Periodic peaks in conductance vs gate voltage

Meaning:

- Each peak = **addition of one electron**

Clear signature of **single-island SET**

7.3 Temperature Dependence

- Oscillations visible up to ~12 K
- At this temperature:
- $kBT \approx 1 \text{ eV}$

Consistent with charging energy (~4 meV)

8. Coulomb Diamonds (Important Analysis)

These are plots of:

- Differential conductance dI/dV
- vs source-drain voltage VSD and gate voltage VG

Features:

- Diamond-shaped regions → no current
- Outside diamonds → conduction allowed

What they reveal:

- Charging energy
- Capacitances

9. Capacitance Analysis

From experiment:

- Charging energy:
- $EC = e^2 / C = 4 \text{ meV}$
- Total capacitance:
- $C \approx 40 \text{ aF}$
- Gate capacitance:
- $CG \approx 10 \text{ aF}$

Comparison with theory:

- Parallel plate estimate:
- $CP \approx 100 \text{ aF}$

Measured value is **2–3× smaller**

→ Due to geometry and quantum effects

10. Effect of Nanowire Diameter

For thicker wire (65 nm):

- Larger capacitance
- Smaller charging energy ($\sim 2 \text{ meV}$)
- Much higher current ($\sim 150\times$)

Reason:

- Larger area → higher capacitance
- Thinner effective barriers

11. Multi-Island Behavior

At **negative gate voltages**:

- Nanowire gets depleted
- Multiple islands form unintentionally

Effects:

- Irregular Coulomb diamonds
- Larger blockade region

Shows transition from:

- **Single-island SET → multi-island system**

12. Electron Density Estimation

- ~ 50 Coulomb diamonds observed
- Corresponds to ~ 50 electrons in island

Estimated electron density:

$\sim 10^{17} \text{ cm}^{-3}$

13. Absence of Negative Differential

Resistance

Why?

- Energy level spacing is small (large dot)

Contrast:

- Smaller dots (~15 nm) → strong quantum confinement
- Can show **resonant tunneling & negative resistance**

14. Key Advantages of This Approach

1. Bottom-up fabrication

- No complex lithography needed

2. Reproducibility

- Device behavior determined by structure, not contacts

3. Scalability

- Suitable for large-scale integration

4. Design flexibility

- Can engineer barriers during growth

15. Limitations & Considerations

- Requires **low temperatures**
- Contact resistance must be minimized
- Gate coupling is not ideal (affects whole wire)

1. What Each Figure Shows (Clear Interpretation)

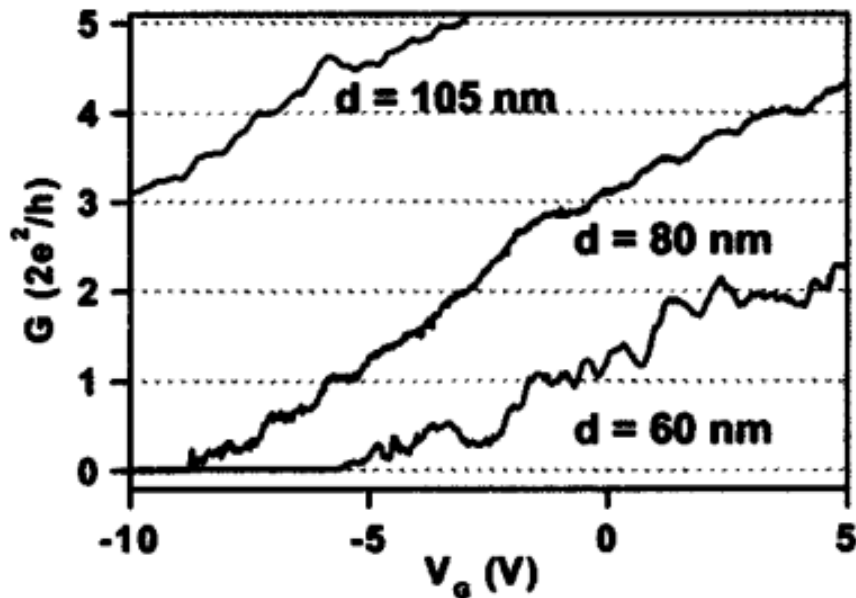


FIG. 1. Conductance plotted as a function of backgate voltage for homogeneous InAs nanowires with diameters of 105, 80, and 60 nm at $T=4.2$ K, showing the depletion of conduction electrons for negative gate voltages. The 105 and 80 nm plots were recorded using a two-terminal lock-in technique with an ac bias of 0.1 mV, and the 60 nm plot with a dc bias of 0.36 mV.

Fig. 1 – Conductance vs Gate Voltage (Homogeneous InAs nanowires)

What is plotted:

- Conductance G vs gate voltage V_G
- For different diameters (105, 80, 60 nm)

Key observations:

- No oscillations → **no Coulomb blockade**
- Conductance decreases at negative V_G → **carrier depletion**
- Thinner wires → higher resistance

Meaning:

- These wires behave like **normal conductors**
- Confirms: **SET effect comes from InP barriers**, not wire itself

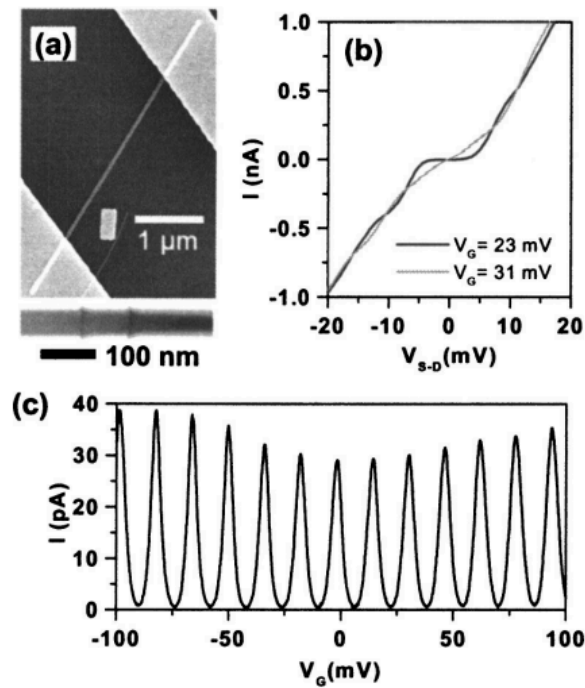


FIG. 2. (a) Scanning electron micrograph of an electrically contacted InAs nanowire with an InP double-barrier structure. The lower inset shows how a selective InAs etch can be used to reveal the InP double barrier (transmission electron micrograph). (b) I - V curves for the 55 nm diameter device in Fig. 2(a) recorded for two different gate voltages at $T=4.2$ K, corresponding to a blocked (black) and a conducting state (gray). (c) Current through the device plotted as a function of the universal back-gate voltage for $V_{\text{SD}} = 0.5$ mV.

Fig. 2(a) – SEM Image of Device

- Shows nanowire with metal contacts
- InP barriers are not visible directly but exist inside

Purpose:

- Confirms physical structure of fabricated SET

Fig. 2(b) – I-V Characteristics

- Current vs source-drain voltage for two gate voltages

Key observation:

- One curve $\rightarrow$ **blocked (Coulomb blockade)** : black
- One curve $\rightarrow$ **conducting** : gray

Meaning:

- Gate voltage controls whether electrons can tunnel

Fig. 2(c) – Current vs Gate Voltage

Key observation:

- **Periodic peaks in current**

Meaning:

- Each peak = **one electron added to island**
- This is the **signature of SET**

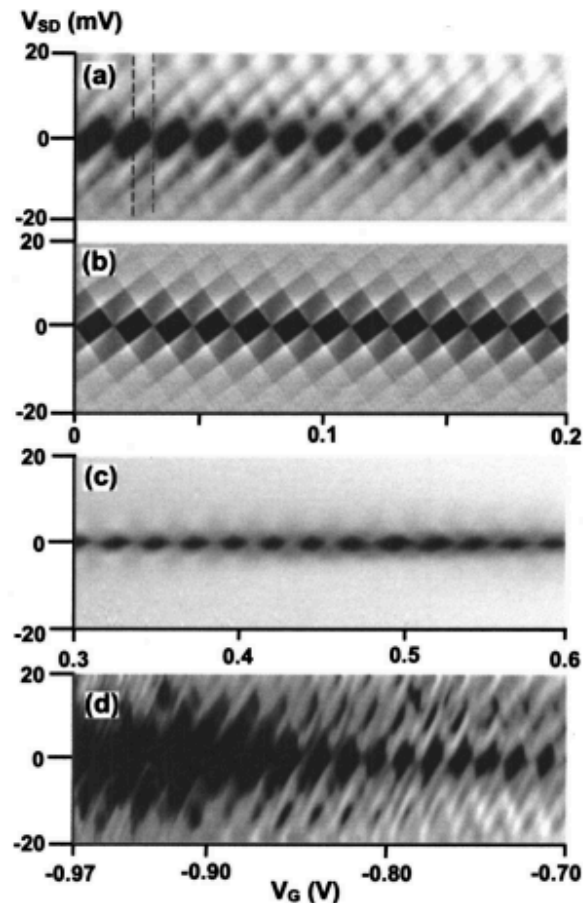


FIG. 3. dI/dV_{SD} plotted as a function of V_G and V_{SD} . Bright areas correspond to a higher value of dI/dV_{SD} . (a) Measurement at $T=4.2$ K for a 55 nm diameter device. The two vertical lines indicate the origin of the $I-V$ curves shown in Fig. 2(b). (b) Theoretical fit at $T=0$ K. (c) Measurement at $T=4.2$ K for a 65 nm diameter device for which the charging energy is reduced by a factor of 2. (d) Depletion of the nanowire in Fig. 3(a) for negative gate voltages, transforming the designed double-barrier device into a multi-island system.

Fig. 3(a) – Coulomb Diamonds (Experimental)

Plot:

- dI/dV vs V_{SD} and V_G

Features:

- Diamond-shaped dark regions → **no current**
- Outside → conduction

Meaning:

- Shows **charging energy**
- Defines allowed/forbidden transport regions

Fig. 3(b) – Theoretical Fit

- Simulation using SET theory

Meaning:

- Matches experiment → validates model

Fig. 3(c) – Larger Diameter Wire

- Diamonds smaller

Meaning:

- **Charging energy reduced**
- Capacitance increased

Fig. 3(d) – Multi-Island Regime

- Irregular diamonds

Meaning:

- Wire breaks into multiple quantum dots at negative V_G

2. Why Current is Periodic with Gate Voltage

Key concept:

Gate voltage changes island energy:

$$E(N) = (N e - C_G V_G)^2 / 2C$$

Condition for conduction:

$$E(N) = E(N+1)$$

Happens periodically → **periodic current peaks**

3. Energy Band Diagram

(A) Zero Electron State

- Island has no extra electron
- Energy levels empty
- Coulomb blockade present

Diagram:

- Two barriers (InP)
- Flat Fermi levels in leads
- No available state in island within bias window

(B) One Electron Addition (First Jump)

Condition:

$$eV \geq e^2 / C$$

What happens:

- First electron tunnels into island
- Energy level aligns with source

Diagram:

- One discrete level in island aligns with source-drain window

(C) After One Electron Enters

Effect:

- Island charge increases $\rightarrow$ energy increases

$$EC = e^2 / C$$

Diagram:

- Energy level shifts upward (Coulomb repulsion)

(D) Second Electron Addition

Condition:

$$E \geq 2 \times e^2 / C$$

Meaning:

- Second electron needs **extra energy**
- Harder to add than first

Diagram:

- Next level shifted even higher

Do R and C change with current?

Answer:

- **Capacitance (C):** constant (geometry dependent)
- **Resistance (R):** approximately constant, but:
 - Effective tunneling rate depends on voltage
 - So current changes due to **energy alignment**, not R/C change

5. SET Theory

Current flows when:

$\mu_{\text{source}} > \mu_{\text{island}} > \mu_{\text{drain}}$

Otherwise:

- Coulomb blockade